

Andrew Park

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EDUCATION

Duke University - Trinity College of Arts and Sciences

Durham, NC

B.S in Mathematics; Secondary Major in Computer Science; Minor in English

Expected May 2028

- GPA: 3.96/4 | First-Year Julia Dale Prize, Duke's highest first-year award for mathematical excellence
- Graduate Coursework: Real Analysis I, Algebraic Structures I, Mathematical Finance, Financial Derivatives
- Tentative: Measure & Integration, Numerical Linear Algebra, Theory & Algorithms of Machine Learning
- CS Coursework: Data Structures & Algorithms, Computer Architecture, Graphics Software Architecture

Stuyvesant High School: ACT 36; PSAT National Merit Scholar

New York, NY

ACADEMIC EXPERIENCE

PRUV Fellow & Undergraduate Researcher, Duke Mathematics

Durham, NC

Advisor: Professor J. Thomas Beale

April 2025 - Present

- Translated Beale's surface-PDE algorithm into a C++ research application, implementing level-set cut-point generation, primary/secondary point classification, sparse equilibration, and Cartesian finite-difference stencils.
- Built solvers for reaction-diffusion PDEs on closed surfaces; simulated Turing-pattern formation and developed validation tests for Laplace-Beltrami diffusion, Poisson, advection, and shallow-water examples.
- Selected as Duke PRUV Fellow for a six-week full-time research project; extending framework toward shallow-water dynamics on exoplanet geometries and preparing Sept. 2026 research talk and thesis/manuscript.

Teaching Assistant & Grader, Duke Mathematics

July 2025 - Present

- Evaluated problem sets for 400+ students across Linear Algebra, Multivariable Calculus, and Differential Equations, providing feedback on proofs and computational methods.

PROJECT EXPERIENCE

Duke Poker Club

Durham, NC

Founder and President

January 2025 - Present

- Re-launched dormant strategy/probability club and grew membership to 100+ students through weekly poker, game theory, and prediction-market events.
- Secured collegiate partnership with Kalshi and trained competitive team for Intercollegiate Poker Association tournaments; interviewed by *The Wall Street Journal* on college poker and prediction markets.

Jupiter: College Basketball Forecasting Model | Python, pandas, scikit-learn

January 2025 - April 2026

- Built a hybrid Elo/PageRank forecasting model incorporating home-court advantage, possession efficiency, and schedule-strength features to estimate NCAA game probabilities.
- Backtested on the 2025 season, achieving 73% out-of-sample accuracy and 0.52 log loss; evaluated probability calibration and error by conference/seed tier.
- Ran Monte Carlo tournament simulations to estimate seed distributions and team advancement probabilities for the 2026 March Madness bracket.

U.S. Election Forecasting Engine | Python, SQL, scikit-learn

September 2021 - June 2024

- Led 7-person team to collect and clean county/precinct election returns from 1912–2024, building SQL/Python pipelines for model-ready voting and demographic datasets.
- Modeled local political realignment with regression and K-nearest neighbors; evaluated forecasts across held-out cycles and geographic segments.

Google Mentorship Program

September 2022 - June 2023

- Built MTA delay-probability model using MTA API; presented architecture weekly to Google engineering mentors.

ADDITIONAL INFORMATION

Technical: C++, Python, SQL, Java, Git, LaTeX; NumPy, pandas, scikit-learn

Relevant Methods: Numerical PDEs, finite differences, Monte Carlo simulation, regression, KNN, backtesting

Languages: English/Korean native; French fluent

Interests: Classical piano, violin (performed on 2023 Grammy-winning orchestral recording), poker, Romanticism, college sports